ARTIN-REES FOR A COHERENT SHEAF AND A QUASI-COHERENT SUBSHEAF ON A NOETHERIAN ALGEBRAIC SPACE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- An affine etale cover of a quasi-compact algebraic space. Properties of Spaces, label spaces-properties-lemma-quasi-compact-affine-cover (invoked at proof line 2378). It provides the affine scheme U with a surjective etale morphism $U \to X$, over which the asserted equality can be checked.
- The source of a morphism to a Noetherian space, locally of finite type, is Noetherian. Morphisms of Spaces, label spaces-morphisms-lemma-locally-finite-type-locally-noetherian (invoked at proof line 2380). It makes U a Noetherian scheme, so that Artin-Rees for schemes is available on it.
- The statement for schemes. Cohomology of Schemes, label coherent-lemma-Artin-Rees (invoked at proof line 2384). It supplies the constant c on the Noetherian scheme U; the same constant works on X because the equality is checked after restriction.

Source reference. Original lemma 01YA: sources/stacks-chow/tags.txt, source lines 2549–2549. Navigation pointer for \ref{coherent-lemma-Artin-Rees} used inside the counted span (line 2385); the argument that consumes it is retained in the span.

Source reference. Original lemma 04ZK: sources/stacks-chow/tags.txt, source lines 6468–6468. Navigation pointer for \ref{spaces-morphisms-lemma-locally-finite-type-locally-noetherian} used inside the counted span (line 2382); the argument that consumes it is retained in the span.

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Source reference. Original lemma 03H6: sources/stacks-chow/tags.txt, source lines 4555–4555. Navigation pointer for \ref{spaces-properties-lemma-quasi-compact-affine-cover} used inside the counted span (line 2379); the argument that consumes it is retained in the span.

Lemma 0.1 (Artin-Rees)**.** *Let S be a scheme. Let X be a Noetherian algebraic space over S. Let $\mathcal{F}$ be a coherent sheaf on X. Let $\mathcal{G} \subset \mathcal{F}$ be a quasi-coherent subsheaf. Let $\mathcal{I} \subset \mathcal{O}_X$ be a quasi-coherent sheaf of ideals. Then there exists a $c \geq 0$ such that for all $n \geq c$ we have*

$$\mathcal{I}^{n-c}(\mathcal{I}^c\mathcal{F} \cap \mathcal{G}) = \mathcal{I}^n\mathcal{F} \cap \mathcal{G}$$

Proof. Choose an affine scheme U and a surjective étale morphism $U \to X$ (see Properties of Spaces, Lemma 03H6). Then U is a Noetherian scheme (by Morphisms of Spaces, Lemma 04ZK). The equality of the lemma holds if and only if it holds after restricting to U. Hence the result follows from the case of schemes, see Cohomology of Schemes, Lemma 01YA. □